

Exp No: 24**MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC****AIM:**

To understand and implement a clock-driven process using

Program:

```
#include <systemc.h>

SC_MODULE(ClockExample) {
    sc_in<bool> clk;

    void process() {
        while (true) {
            wait();
            std::cout << "Time: " << sc_time_stamp()
            << " - Clock Triggered" << std::endl;
        }
    }

    SC_CTOR(ClockExample) {
        SC_THREAD(process);
        sensitive << clk.pos();
    }
};

int sc_main(int argc, char* argv[]) {

    sc_clock clock("clock", 1, SC_NS);

    ClockExample module("module");
    module.clk(clock);

    sc_start(5, SC_NS);
```

```
return 0;
```

```
}
```

The screenshot displays the EDA Playground web interface. At the top, there's a navigation bar with 'New', 'Run', and 'Save' buttons. Below this, the interface is divided into three main sections:

- Left Panel (Tools & Libraries):** Contains settings for the testbench, including the language (C++), libraries (SystemC 2.3.3), and compile options. It also has checkboxes for various run options like 'Use run.bash shell script' and 'Open EPWave after run'.
- Center Panel (Code Editor):** Shows two files: 'testbench.cpp' and 'design.cpp'. The 'design.cpp' file contains the following C++ code:

```
12 cout << "Time: " << sc_time_stamp() << " - Clock Triggered" << endl;
13 }
14
15
16
17 SC_CTOR(ClockExample) {
18     SC_THREAD(process);
19     sensitive << clk.pos();
20 }
21
22
23 int sc_main(int argc, char* argv[]) {
24     sc_clock clock("clock", 1, SC_NS);
25
26     ClockExample module("module");
27     module.clk(clock);
28
29     sc_start(5, SC_NS);
30
31     return 0;
32 }
33
```
- Bottom Panel (Terminal):** Displays the output of the compilation and simulation. It shows the command used to compile the code and the resulting output, which includes the time of each clock trigger.

```
[2026-08-25 08:18:09 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
Done
```